Homework Exercises 6

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After completing all lab exercises below, zip all your solution files into a single archive with your student ID (e.g. `u210023.zip`) and upload it to eClass before the deadline.

Note, you do not have to submit other lab materials such as lab slides or examples.

1 Create Your Own Homepage in Web

1.1 Create an HTML document

Create a page named `index.html` that describes you. (If you are using a lab computer, you can do this using Sublime Text). On your page, include some or all of the following information:

- Your name
- A description of yourself in two sentences or less. Emphasize the most important word(s) by putting them in bold.
- A list of classes you are taking right now at IUT.
- Your 3 favorite movies, books, or TV shows, in order. Make at least one link to an interesting site about that tv show/movie/book, such as its IMDB page.
- Two images, one that represents you when you're happy and the other to represent you when you're sad. (These can be any images you like. Consider searching for images on Google Image Search)
- Something interesting about one or more of your neighbors (people sitting at computers next to you)

For example, the following page in Figure 1 describes Victoria Kirst:

1.2 Format an HTML document

Create a stylesheet named `styles.css` to improve the appearance of your page. Your stylesheet should do the following without any modification to your HTML code:

- Change the color of at least two elements
- Change the font properties of at least two elements (such as family, size, weight, style). Here are some standard fonts you may want to use: Arial, Arial Black, Verdana, Trebuchet MS, Georgia, Tahoma, Courier New, Times New Roman
- Change at least one other thing of your choosing (such as background color, text alignment, etc.)

For example, Figure 2 is Victoria's styled version of her page.

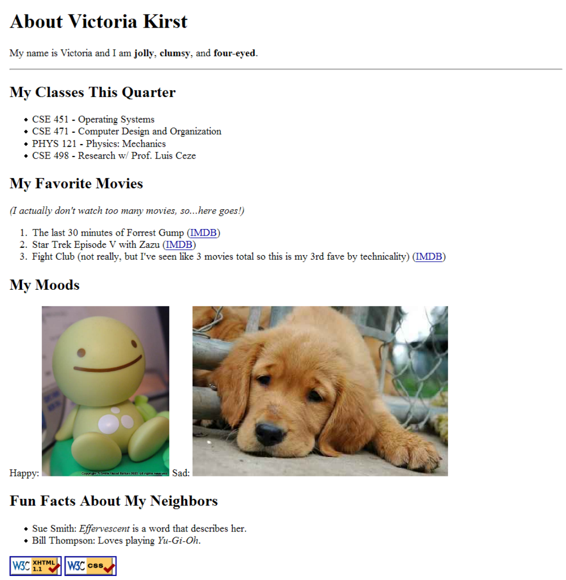

Figure 1: About Me

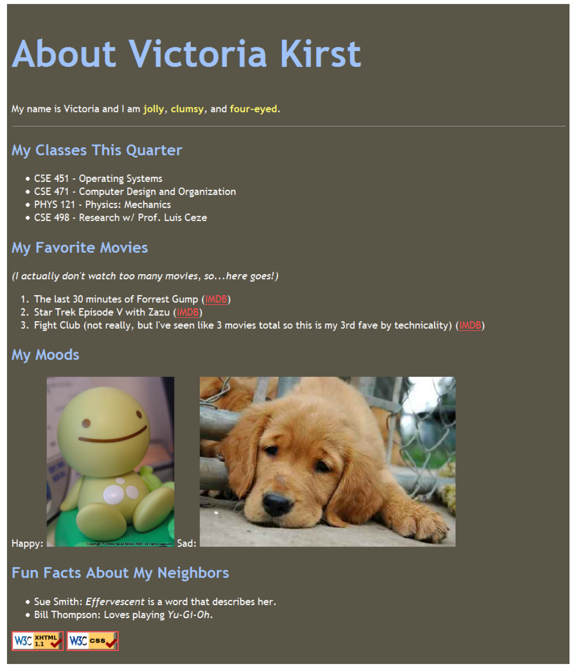

Figure 2: About Me With CSS

1.3 Upload Your Page to Web

Install Git to you computer using this link. Git is the commonly used version-control software, and as a web programmer you will definitely need it in future. You can learn more about Git from its documentation. You can check if Git is successfully installed to your computer by executing following command in your CLI: `git --version`. It should return the currently installed version of Git.

After successfully installing Git on your computer, follow this link and register to GitHub. Then create a new repository[1] and set its name according to this format `YourGithubUsername.github.io`. For example, if your username is `ahmad1994`, then your repository name should be `ahmad1994.github.io`. This is very important, so make sure that you write everything exactly as described above.

Then follow the instructions given on `https://pages.github.com/`. Namely, you need to create a folder where you want to download the contents of the repositorary you have just created in GitHub. For that, open a CLI window [2]. Execute following commands:

```
git clone https://github.com/YourGithubUsername/YourGithubUsername.github.io
```

If your username is `ahmad1994`, then it is

```
git clone https://github.com/ahmad1994/ahmad1994.github.io
```

This will copy the contents of your newly created repository to the designated folder in your computer. Because you haven't put anything to it yet, it is empty now. Now you copy and paste all your HTML, CSS and image files from above tasks into this folder. Once you finish copying, execute the commands below:

```
git add --all
git commit -m "Initial commit"
git push -u origin master
```

[1] You should be able to find *New Repository* button quite easily, once you login to your GitHub profile.

[2] Note, that if you open command line by calling `cmd` in Run, it opens a command line window in your base User folder. You have to change to the right folder. You can do so by using `cd` command.

It will ask for your GitHub username and password. Make sure that you enter them correctly, and press enter[3]. Once you finish everything, you should be able to access your homepage in following url:

```
http://YourGithubUsername.github.io
```

For example, if your username is `ahmad1994`, then it should be as given below:

```
http://ahmad1994.github.io
```

Important: If you complete this task properly, you MUST include your `github.io` URL inside your `index.html`.

2 Python Tasks

Watch **Week 4: Good Programming Practices** lecture videos on this course. Then complete following Python tasks.

1. Run a Python program below to find the index position of the first occurrence of some letter in a given text. For example, if I enter a word `sarvar`, and ask the program to find letter `a`, it should return 1 (note indexing letters starts from 0, and 1 means actually means the second letter). Save your program as `ps2-1.py`.

```
def findIndex(word, letter):
    for i in range(len(word)):
        if word[i].lower() == letter.lower():
            return i
    return -1

word = input('Enter your word')
letter = input('Enter the letter')

print('Position of the letter is: ', findIndex(word, letter))
```

2. Run a Python program below to find the Max of numbers given in the list. Save your program as `ps2-2.py`.

```
def max_num_in_list(list):
    max = list[0]
    for a in list:
        if a > max:
            max = a
    return max

print(max_num_in_list([1, 2, -8, 0]))
```

3. Write a Python function which takes as an argument a list, and returns the number of distinct elements in a given list. For example, if the list is `[2,1,2,3,1,9]`, the number of distinct elements in the list is 4. Save your program as `ps2-3.py`.

4. Write a Python function which takes as an argument a list, and returns a dictionary which gives the frequency of the elements in a list. For example, if the list is `[2,1,2,3,1,9]`, the result should be `{0:0, 1:2, 2:2, 3:1, 4:0, 5:0, 6:0, 7:0, 8:0, 9:1}`. Save your program as `ps2-4.py`.

5. Write a Python function which takes as an argument a list, and returns its reverse as a result. Save your program as `ps2-5.py`.

6. Write a Python function which right rotates a given list by n number of times. For example, right rotation of the list `[2,1,2,3,1,9]` by 2 is `[1,9,2,1,2,3]`. Save your program as `ps2-6.py`.

[3]Note, while entering password in console, you might not see any characters being typed, but it is OK. This is how most consoles mask passwords.